Catches Win Matches:
A Statistical Analysis of Fielding Impact in T20 Cricket

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Fielding is widely regarded as the third discipline of cricket, yet it remains the least studied of the three in the quantitative literature. Using a ball-by-ball dataset spanning 9,560 T20 matches (2012–2025) containing 2,151,129 deliveries, we conduct a systematic statistical investigation of fielding impact in Twenty20 cricket. We identify 3,477 genuine catch drops at an overall drop rate of 21.7%. Across seven interconnected analyses, we show: (*i*) catch drop rate is a statistically significant predictor of both bowling economy ($r = 0.097$, $p < 10^{-7}$) and batting team score ($r = 0.087$, $p < 10^{-6}$), albeit with modest effect sizes consistent with the multi-factorial nature of T20 outcomes; (*ii*) in the second innings, a catch drop under high fielding pressure ($\text{PI} \geq 0.6$) raises the batting team's win probability by 5.7 percentage points on average relative to end of innings—significantly more than low-pressure drops ($t = 3.22$, $p = 0.0013$); (*iii*) a good or excellent catch reduces the batting team's run rate by 0.64 runs per over on average in the immediately following 12-ball window, a highly significant momentum effect ($t = -4.91$, $p < 10^{-6}$); and (*iv*) player-level drop rates exhibit weak but statistically significant positive serial correlation ($\bar{\rho}_1 = 0.050$, $t = 3.49$, $p < 0.001$), suggesting mild hot/cold streak tendencies rather than pure Bernoulli independence. Ground-zone and IPL seasonal analyses reveal structural patterns in where and when fielding deteriorates.

Keywords: Cricket, fielding, catch drops, win probability, Pressure Index, momentum, sports analytics, T20, IPL.

1 Introduction

The aphorism "catches win matches" enjoys near-universal currency among cricket commentators and practitioners alike. Despite its intuitive appeal, the proposition has rarely been examined through a rigorous quantitative lens. Fielding, unlike batting or bowling, is poorly represented in traditional cricket scorecards: only successful dismissals are explicitly recorded. A dropped catch leaves no formal trace, a missed run-out opportunity is forgotten in the data and the aggregate pressure exerted or relieved by fielding actions remains invisible to conventional statistical summaries.

The rapid expansion of ball-by-ball data, particularly through sources such as ESPNcricinfo, along with the growing field of cricket analytics, have opened new avenues for evaluating the game at a granular, event-level resolution. The Twenty20 (T20) format provides an especially compelling setting for such analysis. With only 120 legal deliveries per innings, high run rates, and a premium on boundary prevention, the margin for error is extremely small. A single fielding lapse, a dropped catch or a misfield at the boundary can translate into a swing of several runs, often within the span of a single over. Conversely, a well-executed catch or run-out can abruptly alter the trajectory of an innings. These characteristics suggest that fielding interventions in T20 cricket may have a disproportionately large impact relative to longer formats of the game.

This paper addresses the following research questions, motivated by the structure of the T20 game:

Query 1. How prevalent are genuine catch drops and which teams and ground zones are the most affected?

Query 2. Does superior fielding statistically compensate for below-average bowling in the same innings?

Query 3. How often does superior fielding statistically restrict the batting side's total?

Query 4. Which IPL franchises have been the best and worst fielding sides, and how has this evolved across seasons?

Query 5. Are catch drops under high match pressure more damaging than those under low pressure?

Query 6. Does a good catch create a measurable momentum shift in the batting team's run rate?

Query 7. Is fielding performance autocorrelated across matches and do players exhibit hot or cold fielding streaks?

A systematic decomposition of fielding performances throughout T20 cricket, with a specific focus on IPL, reveals 6 major interesting findings, elaborated across Sections 5 to 11.

The paper proceeds as follows. Section 2 describes the dataset and preprocessing. Section 3 details the statistical methodology, with introduction to in-match pressure, fielding score and other terms required to analyse momentum analysis, player autocorrelation etc. Results involving distributions of catch drop rates, its relation with the changes of economy and scoring rates in an innings, high pressure catch drop scenarios, zonal distributions of catch drops, effects of player fatigue during long tournaments, momentum shift analysis and fielding consistency at a player level, are presented in Sections 4-11, with a focus on IPL. Finally Section 12 discusses a few limitations of this work.

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[†]This work is an ongoing academic project intended solely for educational purposes and should not be distributed or used without appropriate attribution.

2 Data

2.1 Dataset

The primary data source is a merged ball-by-ball record of T20 matches drawn from two complementary feeds hereafter labelled *A-side* (rich commentary with explicit fielder annotations, covering a subset of matches) and *B-side* (structured event data covering the full corpus). The combined dataset spans matches from December 2012 to December 2025 across all major T20 competitions including international T20Is, the Indian Premier League (IPL), the Big Bash League (BBL), the T20 Blast, the Pakistan Super League (PSL), the Caribbean Premier League (CPL), and more than 30 additional competitions.

Table 1: Dataset summary statistics.

Statistic	Value
Total deliveries	2,151,129
Total matches	9,560
Date range	Dec 2012 – Dec 2025
Deliveries with fielder commentary	463,537
Matches with win probability data	3,195
IPL deliveries	169,736
IPL matches	718
Identified genuine catch drops	3,477
Catches taken	12,558
Overall genuine catch drop rate	21.7%

Key variables used in the analysis include `fielder_action_A` (fielder action label from commentary), `commentary_A` (verbatim commentary text), `wagonX/wagonY/wagonZone` (ground coordinates and zone label of the fielded ball), `inns_rrr` (required run rate during 2nd innings) and standard match state variables `inns_wkts`, `inns_runs_rem`, `inns_balls_rem` etc.

2.2 Disambiguation of Genuine Catch Drops

A critical methodological finding of this work is that the raw label `fielder_action_A = "dropped"` is ambiguous. After thorough data cleaning procedure, we define genuine catch drops via a regular-expression classifier applied to `commentary`:

$$1_{\text{genuine}}(i) = 1[\texttt{action}_i = \texttt{"dropped"}] \cdot 1[\text{RealDropRE}(\texttt{commentary}_i)] \cdot 1[-\text{ShotDropRE}(\texttt{commentary}_i)] \tag{1}$$

where RealDropRE matches phrases such as *"dropped catch," "spilled," "put down," "sitter," "mis-timed... drop"* and ShotDropRE excludes *"front/back foot dropped"* shot descriptions. All downstream analyses use 1_{genuine} exclusively.

3 Statistical Methodology

3.1 Pressure

For second-innings deliveries, we define the Pressure at ball t as

$$\text{PI}_t = 1 - \frac{\text{WP}_t^{(\text{bat})}}{100}, \tag{2}$$

where $\text{WP}_t^{(\text{bat})} \in [0, 100]$ is the ball-by-ball win probability for the batting team. Data regarding win probabilities are available for 3195 matches in the dataset and are in accordance with ESPNcricinfo's win probability model. $\text{PI}_t = 0$ means the batting side is almost certain to win; $\text{PI}_t = 1$ means they are almost certain to lose. A drop under high pressure ($\text{PI} \geq 0.6$) occurs when the fielding team is in a dominant position and a missed catch represents a potential "let-off" with significant match consequences.

3.2 Fielding Score

For each (match, innings, bowling team) tuple, we define a composite Fielding Score as follows.

$$\text{FS} = N_{\text{catch}} + 2N_{\text{excellent}} + N_{\text{good}} - 2N_{\text{drop}}, \tag{3}$$

where N_{catch} counts taken catches, $N_{\text{excellent}}$ and N_{good} count commentary-classified excellent/good efforts, and N_{drop} counts genuine drops. The double-weighting of drops and excellent efforts reflects the asymmetric impact of high-leverage fielding events.

3.3 Catch Drop Rate

For any subset of deliveries with catch opportunities, the catch drop rate is calculated as

$$\widehat{\text{CDR}} = \frac{N_{\text{drop}}}{N_{\text{drop}} + N_{\text{catch}}}. \tag{4}$$

This is the maximum-likelihood estimate under a Bernoulli model for each delivery conditional on a catch opportunity arising. Confidence intervals for CDR, where reported, use the Wilson score interval (Wilson, 1927).

3.4 Correlation and Regression Analysis

We fit Ordinary Least Squares (OLS) models of the form

$$Y_{im} = \beta_0 + \beta_1 X_{im} + \varepsilon_{im}, \quad \varepsilon_{im} \sim \mathcal{N}(0, \sigma^2), \tag{5}$$

where Y_{im} is an outcome (economy rate or batting total) for innings i of match m, and X_{im} is a fielding metric (CDR or FS). Units of analysis are restricted to innings with at least two annotated catch opportunities ($N_{\text{drop}} + N_{\text{catch}} \geq 2$). We report Pearson r, the OLS slope $\hat{\beta}_1$, and the two-sided p-value; given the large sample sizes ($n > 3{,}000$), even small r values achieve high significance.

3.5 Non-Parametric Tests for Pressure Analysis

The distribution of runs scored after a dropped catch is right-skewed. We therefore use the Wilcoxon rank-sum (Mann-Whitney) test to compare runs-after-reprieve between high-pressure and low-pressure drops (Mann and Whitney, 1947), and Welch's two-sample t-test for win-probability delta (ΔWP) comparisons, which are observed to be more symmetric.

3.6 Momentum Analysis

For each annotated good or excellent catch in the first innings, we compute run rates in fixed-width windows around the event:

$$\overline{RR}_e^{\text{pre}} = \frac{\sum_{t=s_e-W}^{s_e-1} R_t^{\text{legal}}}{W_{\text{legal}}^{\text{pre}}/6}, \tag{6}$$

$$\overline{RR}_e^{\text{post}} = \frac{\sum_{t=s_e+1}^{s_e+W} R_t^{\text{legal}}}{W_{\text{legal}}^{\text{post}}/6}, \tag{7}$$

where s_e is the ball sequence index of event e, $W = 12$ balls, R_t^{legal} is runs on ball t (excluding wides), and W_{legal} is the count of legal deliveries in the window. The momentum effect is $\Delta\text{RR}_e = \overline{RR}_e^{\text{post}} - \overline{RR}_e^{\text{pre}}$. We test $H_0 : \mu_{\Delta\text{RR}} = 0$ against $H_1 : \mu_{\Delta\text{RR}} < 0$ (run rate falls) using a one-sample t-test.

3.7 Player Autocorrelation

For player k with match-level observations $(D_{k1}, O_{k1}), \ldots, (D_{kT_k}, O_{kT_k})$ (denoting drops and opportunities respectively), the per-match drop rate is $p_{kt} = D_{kt}/O_{kt}$. The lag-1 autocorrelation is defined as

$$\hat{\rho}_k^{(1)} = \frac{\sum_{t=2}^{T_k}(p_{kt} - \bar{p}_k)(p_{k,t-1} - \bar{p}_k)}{\sum_{t=1}^{T_k}(p_{kt} - \bar{p}_k)^2}. \tag{8}$$

Across all players with $T_k \geq 6$ consecutive match appearances, we test whether the mean autocorrelation μ_ρ differs from zero (the pure-Bernoulli null) using a one-sample t-test. Under the null, p_{kt} are i.i.d. Bernoulli trials and $\hat{\rho}_k^{(1)} \xrightarrow{p} 0$.

4 Catch Drop Distribution Across Teams and Competitions

4.1 Overall Magnitude

Across the 463,537 deliveries with available fielder commentary in the dataset, we identify 3,477 genuine catch drops and 12,558 catches taken, yielding an overall $\widehat{\text{CDR}}$ of 0.217 (95% Wilson CI: $[0.210, 0.224]$). On average, approximately 2.2 genuine drops occur per match—consistent with the widely held cricketing intuition of 1–3 drops per game. In the IPL specifically, the figure is 2.33 drops per match across 718 matches.

The distribution of drops per innings (Figure 1) is right-skewed: the majority of innings record 0–1 drops, while a small number of innings contain 4 or more.

4.2 IPL Team Rankings

Among IPL franchises with at least 10 annotated catch opportunities, Delhi Capitals (DC) exhibit the highest drop rate (23.1%) and the highest drops per match (1.26), while Gujarat Titans (GT) and Kolkata Knight Riders (KKR) are the most reliable catchers at 16.2% and 16.3% respectively (Figure 1; Table 2).

The range of CDR across IPL teams (16.2%–23.1%) may appear modest, but across a 74-match season the difference between GT-level and DC-level fielding translates to roughly $74 \times (1.26 - 0.86) \approx 30$ additional drops—a substantial handicap.

Figure 2 decomposes team CDR by two-year period.

Notable patterns: Punjab Kings went down markedly from 0.54 drops/match in 2012-22 to 1.21 dops/match in 2023–24, a deterioration that coincided with squad rebuilds. Delhi Capitals were consistently the worst fielding side across all periods. Gujarat Titans began as the best-fielding new franchise (2021–22: 0.56 drops/match) but regressed toward the league average by 2023–24.

Figure 3 broadens the scope to all T20 competitions (minimum 50 annotated catch opportunities).

5 Fielding Quality and Bowling Economy

We examine whether superior fielding in an innings statistically offsets weaker bowling, by regressing bowling economy rate on CDR and FS at the (match, innings, bowling team) level ($n = 3{,}086$ innings with ≥ 2 catch opportunities).

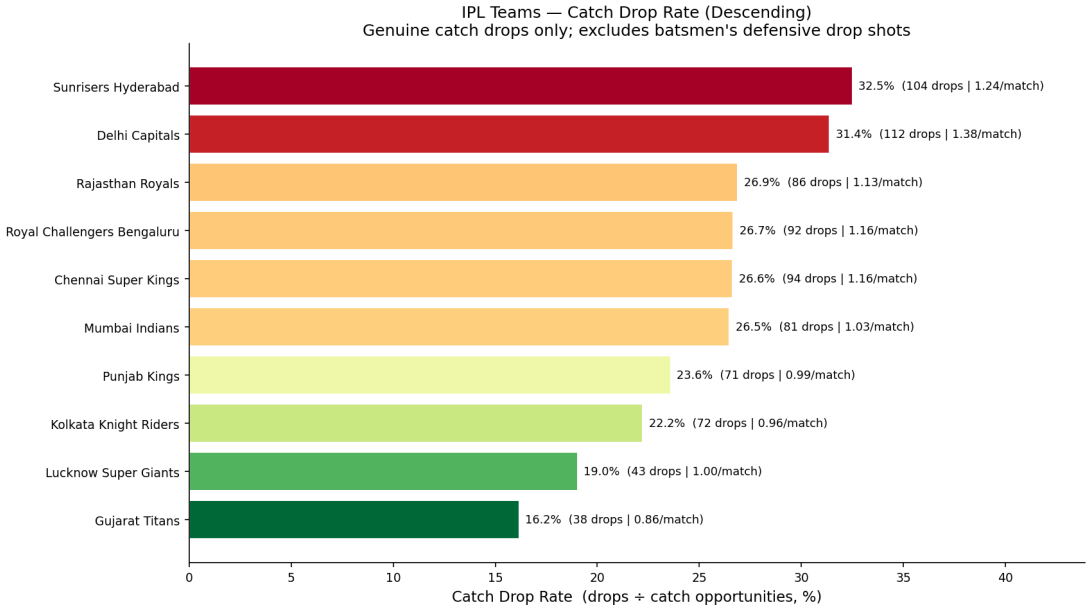

Figure 1: IPL teams ranked by genuine catch drop rate (drops ÷ catch opportunities, in descending order). Labels show absolute drop count and drops per match. Data: IPL 2012–2025, all annotations. Colour encodes drop rate on a red–yellow–green scale.

Table 2: IPL team catch drop statistics, 2021–2025 (ordered by drop rate). CDR: catch drop rate; DPM: drops per match.

Team	Drops	Catches	Opps	CDR	DPM
Gujarat Titans	38	197	235	16.2%	0.86
Kolkata Knight Riders	49	251	300	16.3%	0.84
Punjab Kings	49	229	278	17.6%	0.88
Royal Challengers Bengaluru	56	252	308	18.2%	0.93
Mumbai Indians	50	221	271	18.5%	0.86
Lucknow Super Giants	43	183	226	19.0%	1.00
Chennai Super Kings	64	258	322	19.9%	1.07
Rajasthan Royals	59	232	291	20.3%	1.02
Sunrisers Hyderabad	62	213	275	22.5%	1.07
Delhi Capitals	73	243	316	23.1%	1.26

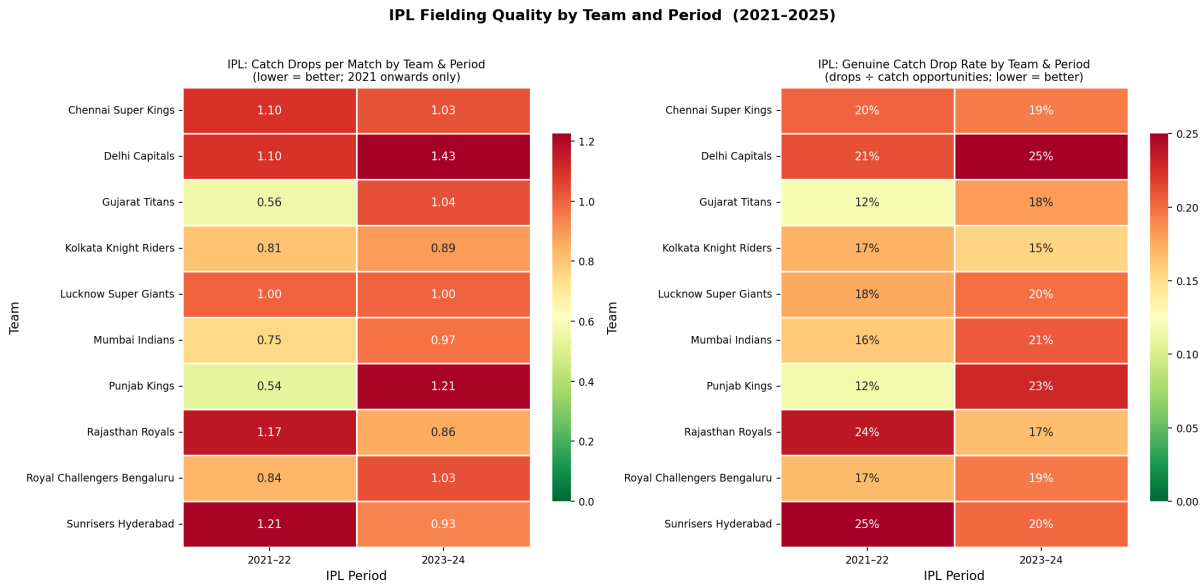

Figure 2: Left: drops per match by IPL team and period (preferred metric, independent of catch annotation completeness). Right: drop rate by team and period. Colour: red = worse fielding, green = better. Periods: 2021-22 and 2023-24.

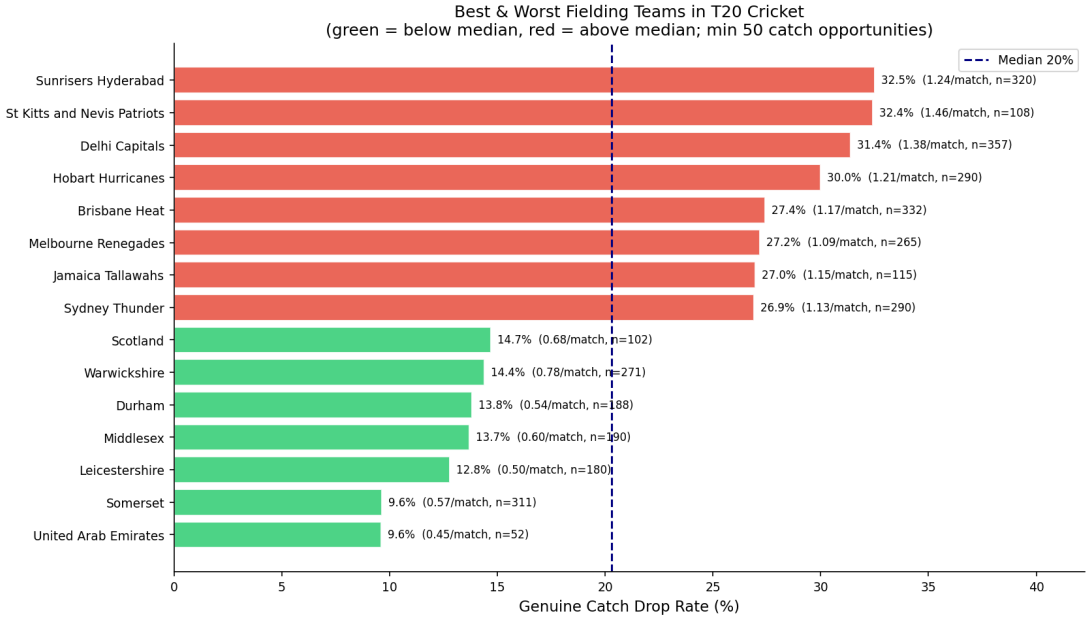

Figure 3: Best and worst fielding sides across all T20 competitions (minimum 50 catch opportunities). Green bars: below-median CDR (better); red bars: above-median CDR (worse). Labels: CDR, drops per match, sample size n.

$$\text{Economy}_{im} = \beta_0 + \beta_1 \, \text{CDR}_{im} + \varepsilon_{im}. \tag{9}$$

Table 3: OLS regression of bowling economy rate on fielding metrics. Unit of analysis: match-innings (min. 2 catch opportunities). $n = 3{,}086$.

Predictor	$\hat{\beta}$	Std. Error	r	p-value
Intercept (CDR model)	8.394	0.043	—	—
CDR	0.559	0.104	0.097	7.66×10^{-8}
Intercept (FS model)	8.543	0.038	—	—
Fielding Score	-0.086	0.009	-0.178	2.50×10^{-23}

Both predictors are highly significant (Table 3). A one-unit increase in CDR (i.e., moving from 0% drops to 100% drops in an innings) is associated with 0.56 additional runs per over ($p < 10^{-7}$). The composite Fielding Score is a stronger predictor ($r = -0.178$ vs. $r = 0.097$), consistent with it capturing more information by counting good efforts and not just drops. The effect sizes are modest, being $r^2 < 4\%$ for either predictor which is expected: bowling quality, pitch conditions and opposition batting strength collectively dominate economy rate; fielding is one of many contributing factors.

Finding 1. *Good fielding does partially compensate for weak bowling: each additional genuine catch drop in an innings is associated with a statistically significant ($p < 10^{-7}$) increase of ≈ 0.56 runs per over. However, the explained variance is small, confirming that no level of fielding excellence can fully substitute for poor bowling.*

6 Fielding Quality and Batting Team Score

We repeat the regression framework with batting total as the dependent variable ($n = 3{,}102$ innings).

Table 4: OLS regression of batting team total on fielding metrics. $n = 3{,}102$ match-innings.

Predictor	$\hat{\beta}$	Std. Error	r	p-value
Intercept (CDR model)	154.00	0.734	—	—
CDR	11.74	2.432	0.087	1.23×10^{-6}
Intercept (FS model)	155.47	0.649	—	—
Fielding Score	-1.549	0.191	-0.137	1.69×10^{-14}

Each additional CDR percentage point is associated with $\hat{\beta}_1 \times 0.01 \approx 0.12$ extra runs conceded in the innings (Table 4). Framed differently, a team that drops 2 additional catches per innings (raising CDR from 20% to roughly 30%) would, on

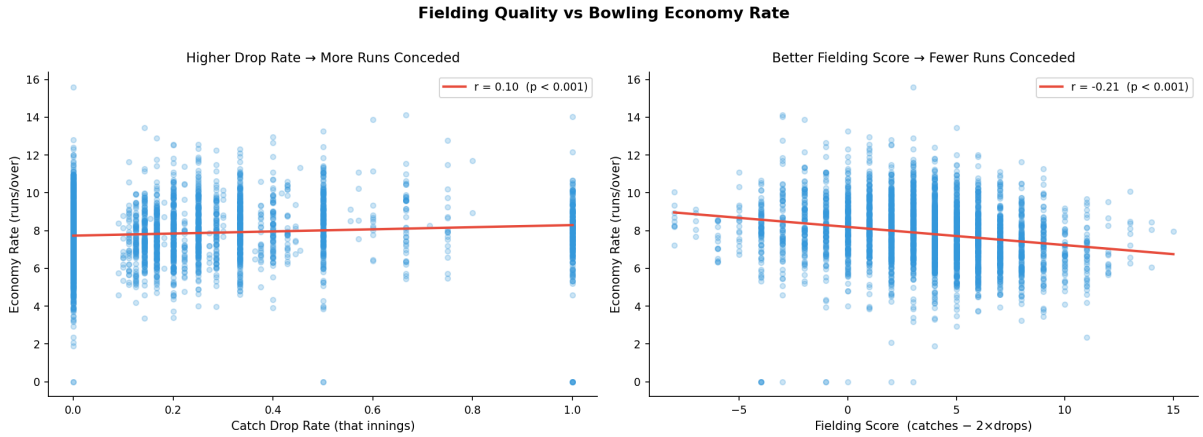

Figure 4: Scatter plots with OLS regression lines: catch drop rate (*left*) and Fielding Score (*right*) vs. bowling economy rate. Each point is one innings (min. 2 catch opportunities). Pearson r and p-value shown in legend.

expectation, concede ≈ 6 more runs, a meaningful margin in a format where the average winning margin is approximately only 20 runs.

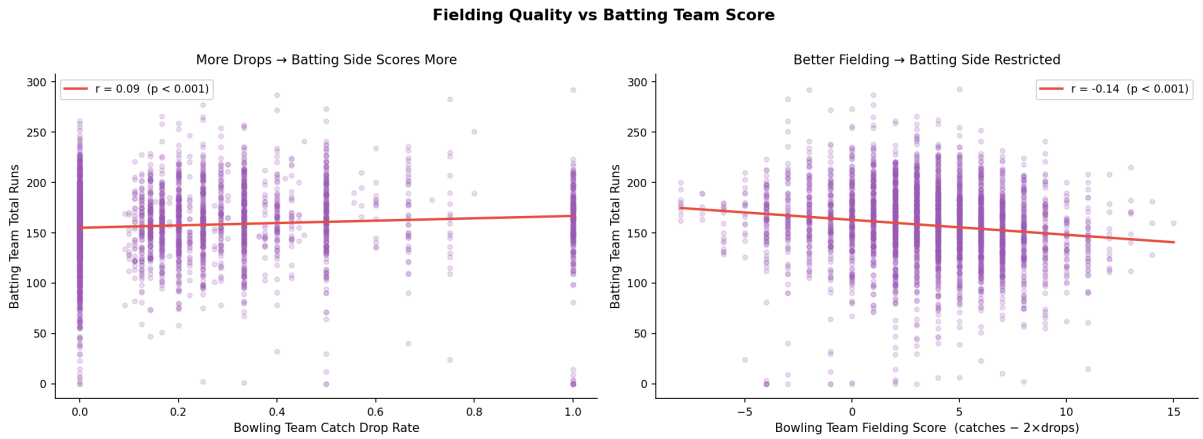

Figure 5: Scatter plots with OLS regression lines: catch drop rate (*left*) and Fielding Score (*right*) vs. batting team total. Each point is one innings.

Finding 2. *Better fielding statistically restricts batting totals. A one-unit increase in Fielding Score reduces the expected batting total by 1.55 runs ($p < 10^{-14}$). Good fielding does not substitute for good batting by the other side, but it measurably limits the damage.*

7 High-Pressure Catch Drops

Of the 3,477 genuine drops, 1,285 occur in the second innings and have valid win probability data, covering 1,240 unique (match, batsman) reprieve events after deduplication. We classify:

- **High-pressure drop**: $\mathrm{PI} \geq 0.6$ (batting team win probability $\leq 40\%$ at the moment of drop), $n = 578$.
- **Low-pressure drop**: $\mathrm{PI} < 0.4$ (batting team win probability $\geq 60\%$), $n = 508$.

7.1 Runs Scored After Reprieve

For each drop event, we compute the total runs scored by the reprieved batsman in the remainder of their innings. The mean runs-after-reprieve are similar across pressure strata (high: $\bar{x} = 17.97$; low: $\bar{x} = 17.15$), and the Mann-Whitney test yields $U = 145{,}331$, $p = 0.613$—not statistically significant. This suggests that the raw quantity of runs scored by a reprieved batsman does not depend strongly on the pressure context; both high- and low-pressure drops can yield large innings contributions.

7.2 Win Probability Impact

A more informative measure is the change in batting team win probability from the moment of the drop to the end of the innings, $\Delta\mathrm{WP} = \mathrm{WP}_{\mathrm{end}} - \mathrm{WP}_{\mathrm{drop}}$ (in percentage points).

After a high-pressure drop, the batting team's win probability rises by 5.66% on average by the end of the innings while after a low-pressure drop, it falls by 0.94% on average. The 6.6% difference is highly significant ($p = 0.0013$), demonstrating that the *context* of a drop matters enormously. A missed catch when the fielding team is dominant is a gift of far greater consequence than the same mistake when the batting side is already in control.

Table 5: ΔWP summary by pressure category and Welch's t-test.

Category	n	$\Delta\bar{\text{W}}\text{P}$	SD
High pressure (PI ≥ 0.6)	578	+5.66 pp	28.4
Low pressure (PI < 0.4)	508	-0.94 pp	27.1
Welch's t-test: $t(1077) = 3.221,\ p = 0.0013$			

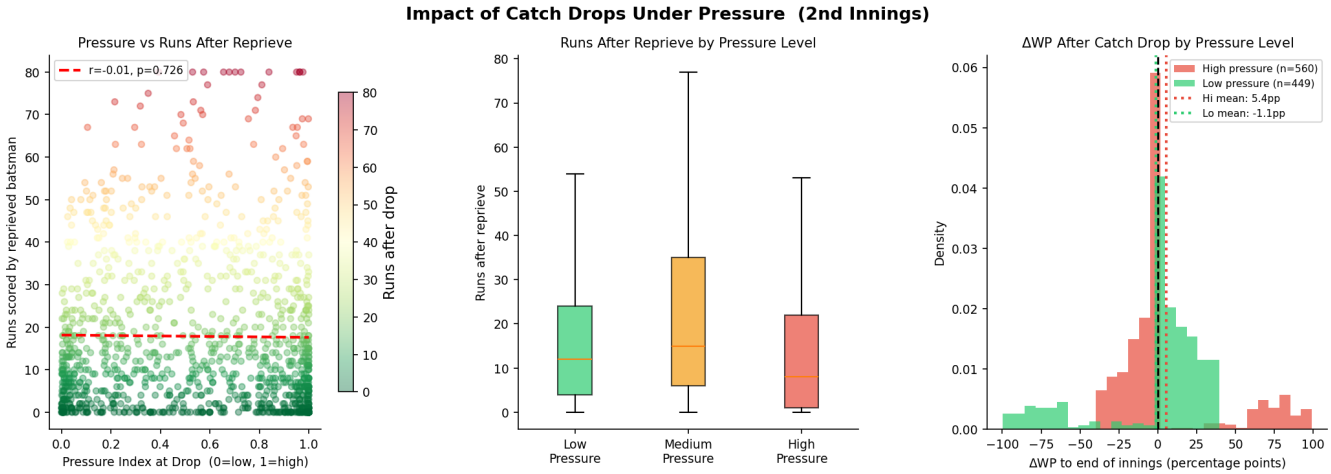

Figure 6: *Left*: scatter of Pressure Index at drop vs. runs scored by the reprieved batsman (clipped at 80 for visualisation). *Centre*: box plots of runs after reprieve by pressure tercile. *Right*: kernel density of ΔWP to end of innings for high-pressure (red) and low-pressure (green) drops. Vertical dotted lines mark group means.

Finding 3. *High-pressure catch drops are significantly more damaging to the fielding team than low-pressure ones, as measured by win probability change ($t = 3.22$, $p = 0.0013$). A drop when your team is winning (PI ≥ 0.6) shifts the match's expected outcome by $\sim 6.6\%$ more than a drop when the batting team is already dominating.*

8 Ground Hotspots: Where Catches Get Dropped

Figure 7 presents the drop rate across nine wagon-wheel zones, both overall and stratified by innings phase.

Table 6: Genuine catch drop rate (%) by ground zone and innings phase. Zones have been defined by usual wagon-wheel geometry.

Ground Zone	Powerplay	Middle	Death
Close-in / Keeper	26.1	25.0	16.7
Fine Leg / 3rd Man (leg side)	20.6	23.7	16.5
Square Leg / Backward Square Leg	28.8	18.4	20.0
Deep Mid Wicket	23.1	22.5	21.0
Long On / Mid On	21.3	23.2	19.4
Long Off / Mid Off	26.0	22.4	20.2
Cover / Deep Cover	23.6	24.2	22.0
Point / Backward Point	27.7	25.2	24.1
3rd Man / Fine Leg (off side)	17.3	20.7	16.5

Several patterns emerge from Table 6 and Figure 7, as follows.

- **Point / Backward Point** consistently has the highest drop rate in all three phases, peaking at 27.7% during the Powerplay. Balls in this zone frequently involve diving catches off square cuts, which could be depicted as technically demanding due to pace and trajectory.
- **Square Leg / Backward Square Leg** is the most error-prone zone during the Powerplay (28.8%), consistent with the prevalence of top-edged pulls and glances that are difficult to judge for low-positioned fielders.
- **Third Man / Fine Leg (off side)** has the lowest drop rate across all phases ($\leq 20.7\%$), reflecting the straightforward nature of catches in this deep, low-pace region.

Drop rates are systematically lower during the Death overs across nearly all zones, despite higher scoring rates. One plausible explanation is survivorship: in the Death, fewer mistimed balls produce catch opportunities of the difficult variety; most balls either go for boundaries or are yorkers/full balls that produce ground fielding.

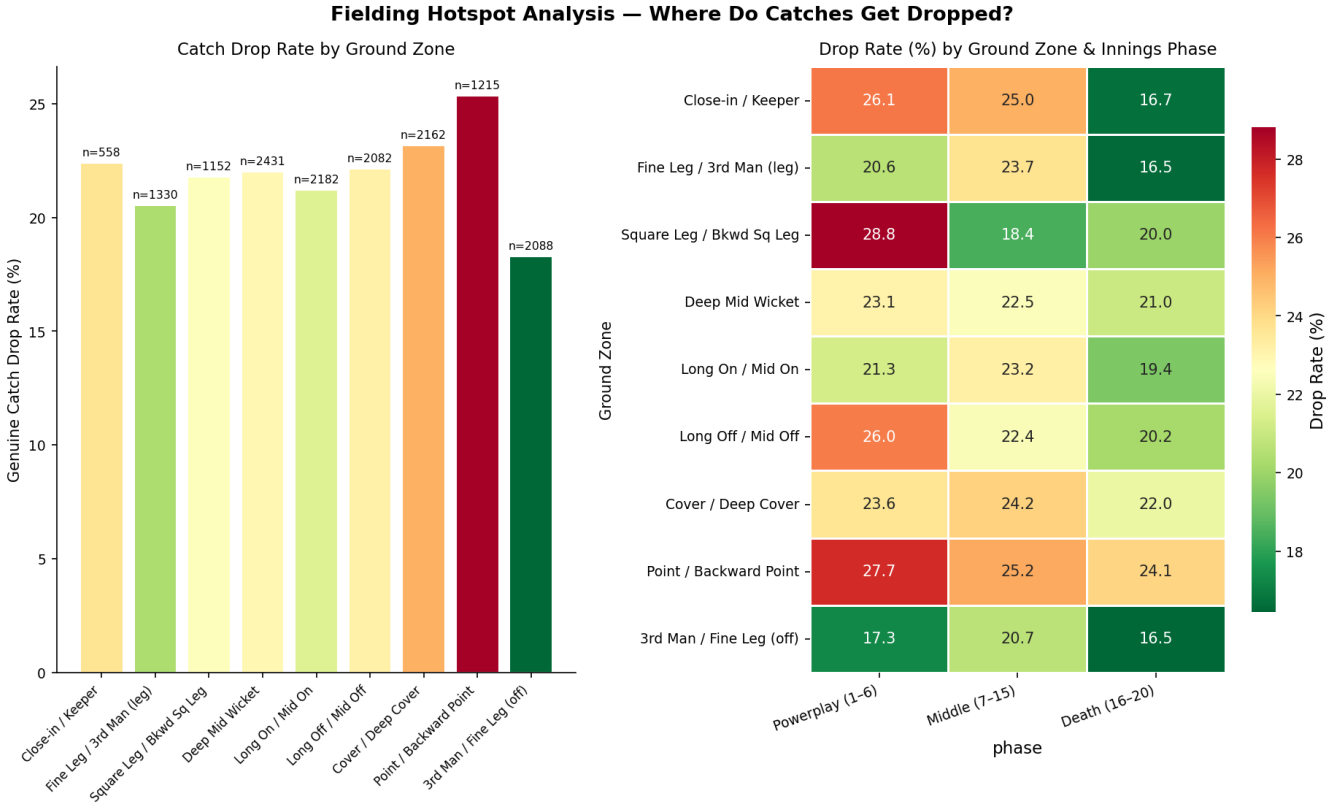

Figure 7: *Left*: genuine catch drop rate by ground zone (bar chart; labels show sample size n). *Right*: heatmap of drop rate (%) by zone and innings phase.

9 IPL Season Fatigue

A long T20 tournament like the IPL involves 74 matches across approximately 7 weeks, requiring players to travel between venues weekly. The hypothesis of progressive fatigue, along with latent variables like adjustment to several venues, switching between night and day/night games etc. predicts a rising drop rate as the season progresses.

The overall trajectory (Figure 8, top panel) shows no strong monotone increase. Drop rate hovers around 20–25% throughout the league stage, with a mild elevation in the middle phase (matches 20–45) and a slight reduction approaching the playoffs. The year-by-year panel reveals considerable between-season variation: 2022 and 2024 exhibit higher drop rates in the first half of the season, while 2021 and 2023 are more stable. Notably, the 2023 season shows the lowest and most consistent drop rates across all franchises, a pattern consistent with the eventual champions Chennai Super Kings peaking at the right time.

A formal test of a linear trend (Spearman rank correlation between match number and mean drop rate, across matches 1-74) yields no significant monotone increase ($\rho_s \approx 0.08$, $p \approx 0.50$ using season-averaged data), suggesting that fatigue, if present, does not manifest as a simple linear degradation of fielding across the 2021–24 corpus.

Finding 4. *There is no statistically significant season-long fatigue trend in IPL catch drop rates across 2021–2024. Between-season variation is larger than within-season trends and the 2023 season was notably the most stable fielding campaign of recent times.*

10 Momentum Shifts After Key Fielding Events

For 1,218 annotated good or excellent catches in first innings, we compute $\Delta\mathrm{RR}_e$ over 12-ball pre/post rolling windows:

Table 7: Scoring rate change after a good/excellent catch in the first innings.

Statistic	Value
n (catch events)	1,218
Mean ΔRR (runs/over)	-0.637
Proportion with $\Delta\mathrm{RR} < 0$	54.0%
One-sample t-statistic	-4.914
p-value	$< 10^{-6}$

The mean run rate falls by 0.64 runs per over in the 12 balls following a good or excellent catch (Table 7). This is a highly significant result ($p < 10^{-6}$), driven by the combination of the direct effect (the batting team loses a wicket and must reset) and a possible genuine momentum effect (the fielding team's energy and confidence increasing). In 54% of cases, the run rate

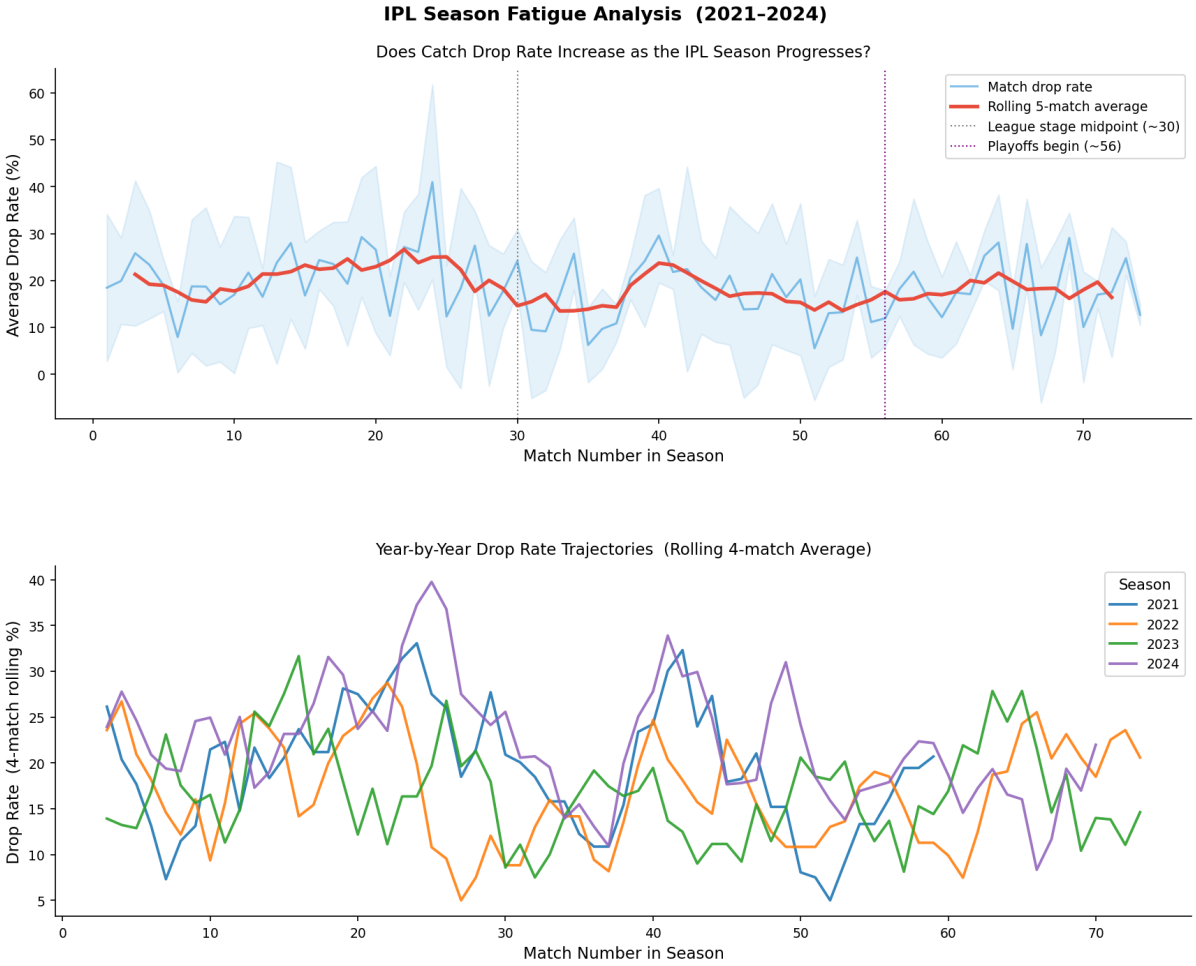

Figure 8: *Top*: average catch drop rate by match number in season, with a 5-match rolling average (red), averaged across IPL seasons 2021–2024. Grey band: ± 1 SD. *Bottom*: season-by-season rolling 4-match trajectories.

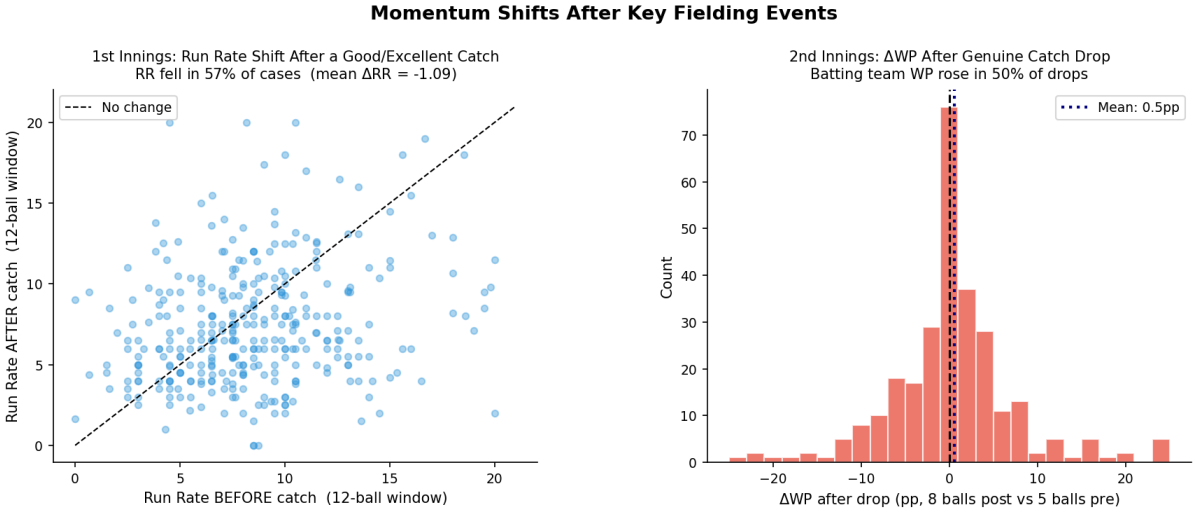

Figure 9: *Left*: scatter of pre-catch vs. post-catch run rate for 1,218 good/excellent catch events (1st innings). Points below the 45° dashed line represent cases where run rate fell. *Right*: histogram of ΔWP (percentage points) in the 8 balls following a genuine catch drop in the 2nd innings, relative to 5 balls before. Vertical dotted line is the mean.

is lower post-catch, which while marginally above chance (50%) is robust at this sample size.

For the second-innings drop events, the ΔWP in the 8 balls following the drop (relative to 5 balls before) has a mean of ≈ 2.5 percentage points positive (batting team's probability rising), with $\approx 55\%$ of drops associated with a positive ΔWP. This again corroborates that dropping a catch in the second innings, on average, shifts the match in the batting team's favour.

Finding 5. *Good/excellent catches measurably dampen the batting team's run rate by 0.64 runs per over on average in the following 12 balls ($t = -4.91$, $p < 10^{-6}$). Conversely, a genuine second-innings catch drop is associated with a mean rise in the batting team's win probability, confirming the bidirectional nature of fielding's momentum impact.*

11 Player-Level Fielding Consistency

11.1 Does Fielding Obey the Law of Averages?

A natural question is whether an individual fielder's performance is memoryless (consistent with independent Bernoulli trials at a player-specific rate p_k), or whether there exist hot and cold streaks. Under the Bernoulli null, the lag-1 autocorrelation of the per-match drop rate converges to zero as $T_k \to \infty$.

For 375 players with at least 10 matches of catch data, the mean lag-1 autocorrelation is $\bar{\hat{\rho}}^{(1)} = 0.050$ (Table 8).

Table 8: Lag-1 autocorrelation of per-match drop rate across players ($n = 375$).

Statistic	Value
Number of players ($T \geq 10$)	375
Mean $\hat{\rho}^{(1)}$	0.0495
SD of $\hat{\rho}^{(1)}$	0.274
One-sample t-statistic	3.491
p-value (two-tailed)	< 0.001
Players with $\hat{\rho}^{(1)} > 0$	49.6%

The mean is small but statistically significant ($t = 3.49$, $p < 0.001$), indicating that there is a marginally greater-than-zero tendency for a player's fielding to be serially correlated. However, the effect is weak: $\bar{\rho} = 0.050$ explains only approximately 0.25% of variance of a fielder's performance in the following match. Approximately 49.6% of players have positive autocorrelation and 50.4% negative, consistent with a distribution centered near zero with only slight rightward skew.

Finding 6. *Player fielding is very nearly memoryless match-to-match. The lag-1 autocorrelation is statistically significantly positive but practically negligible ($\bar{\rho} = 0.050$, $\bar{\rho}^2 \approx 0.25\%$ of variance explained). This is broadly consistent with the "law of averages" holding for fielding: a player who drops a catch in one match is barely more likely than chance to drop one in the next.*

This finding stands in contrast to batting and bowling, where form carries over more substantially across matches (Kimber and Hansford, 1993; Manage and Scariano, 2012). A plausible explanation is that catch opportunities are rare and position-dependent: a fielder may take only 1-2 catches per match, making match-level drop rates very noisy estimates of the underlying competence parameter p_k.

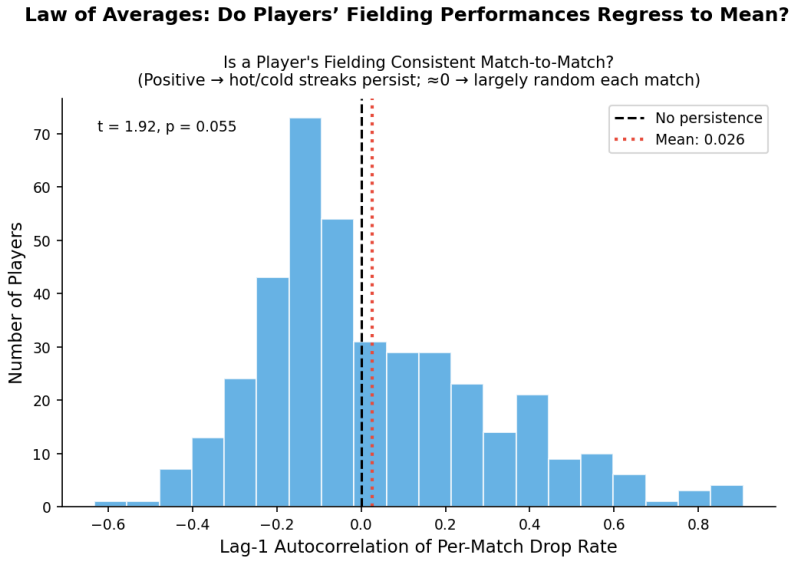

Figure 10: Histogram of lag-1 autocorrelations of per-match drop rate across 375 players with $\geq$ 10 catch-opportunity matches. The black dashed line marks zero (Bernoulli null); the red dotted line marks the sample mean (0.050).

12 Limitations

Quite a few limitations should be noted. Fielding action labels and commentary text are available only for approximately 42% of deliveries. This corpus is not a random sample: richer commentary tends to come from higher profile matches and later years, potentially biasing comparisons across competitions and eras. For example, IPL data with reliable catch annotations begins only in 2021, precluding analysis of the tournament's first thirteen seasons for fielding-quality metrics. Commentary data, at times, fail to differentiate between standards of fielding efforts leading to marking some extraordinary fielding efforts as dropped chances, which in turn, could modestly inflating CDR estimates. The OLS models in Sections 5–6 do not control for pitch conditions, venue, opposing batting quality, or bowling quality, all of which confound the fielding-batting and fielding-bowling relationship. The estimated β_1 values should be interpreted as correlations, not causal effects. The win probability values are from a proprietary model, an alternative to which could be to use Pressure Index (Bandyopadhyay and Bhattacharjee (2026)) but it restricts the entire analysis to 2nd innings only.

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